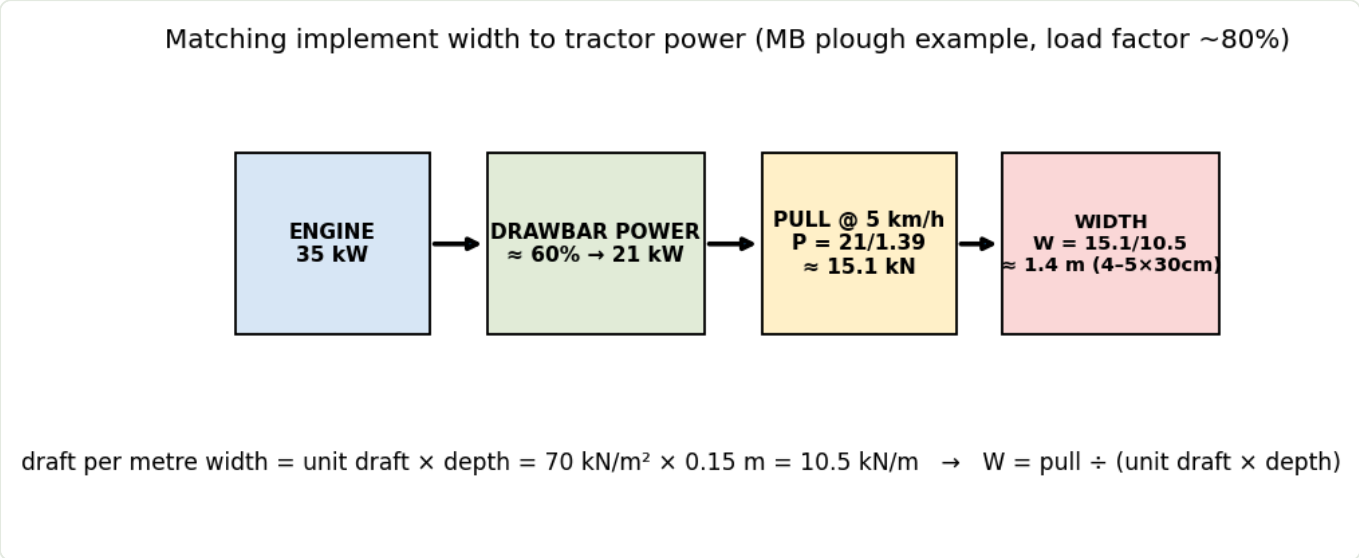
UNIT 3 · TOPIC 10(A)

Matching the Power–Implement System

Syllabus topic: Matching power-implement system; estimation of energy and power requirements

Matching, step by step

1. Available drawbar power = engine power × transmission × tractive efficiency (≈ **50–75%**; Unit 2, Topic 5).
2. Pull available at the chosen speed = power ÷ speed.
3. Implement width the tractor can handle: **W = pull ÷ (unit draft × depth)** for tillage, or **W = pull ÷ draft-per-metre** for width-based implements.



Matching implement width to tractor power (MB plough example)

- Load the tractor to ~**75–85%** of rated power in the selected gear. Overloading stalls/lugs the engine and over-slips the wheels; underloading wastes capacity and fuel.
- For light loads, use "**gear-up, throttle-down**".
- Match PTO-driven machines by **power (kW), not draft** — thresher/rotavator demand vs PTO rating.
- A rotavator partially "pushes" the tractor (**negative draft**), reducing wheel-slip losses.

Worked chain. 35-kW tractor × 60% = 21 kW drawbar; pull at 5 km/h (1.39 m/s) = 21/1.39 = **15.1 kN**; draft per metre = 70 kN/m² × 0.15 m = **10.5 kN/m**; width W = 15.1/10.5 = **1.44 m**.

Practice MCQs

Q126. A 35-kW tractor with 60% power delivery gives drawbar power of:

(a) 15 kW (b) 21 kW
(c) 28 kW (d) 35 kW

Q127. Pull available from 21 kW at 5 km/h (1.39 m/s) is about:

(a) 8 kN (b) 12 kN
(c) 15 kN (d) 21 kN

Q128. With unit draft 70 kN/m² at 15 cm depth, draft per metre width is:

(a) 1.05 kN
(b) 4.7 kN
(c) 10.5 kN
(d) 70 kN

Q129. Hence the matched plough width for 15.1 kN pull is about:

(a) 0.7 m
(b) 1.4 m
(c) 2.8 m
(d) 4.2 m

Q130. For best economy a tractor should run loaded to about:

(a) 30–40%
(b) 50%
(c) 75–85%
(d) 100–110% of rated power

Q131. For light loads, fuel is saved by:

(a) High gear, reduced throttle (gear-up throttle-down)
(b) Low gear, full throttle
(c) Half clutch
(d) Extra ballast

Q133. A rotavator reduces drawbar demand because it:

(a) Has no blades
(b) Generates forward thrust (negative draft) via PTO power
(c) Is lighter
(d) Cuts shallow only

Answers

Q126 — (b) 21 kW — *35×0.6 = 21 kW.*
Q127 — (c) 15 kN — *Pull = 21/1.39 = 15.1 kN.*
Q128 — (c) 10.5 kN — *Per-metre draft = 70×0.15 = 10.5 kN/m.*
Q129 — (b) 1.4 m — *W = 15.1/10.5 = 1.44 m.*
Q130 — (c) 75–85%
Q131 — (a) High gear, reduced throttle (gear-up throttle-down)
Q133 — (b) Generates forward thrust (negative draft) via PTO power

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 2 (Field Capacities and Cost Analysis); Hunt — Farm Power and Machinery Management; Jagdishwar Sahay; SMAM/CHC guidelines; NPTEL lectures.